SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME:DATA ANALYSIS WITH SPSS/SAS/R

PRACTICAL 8

AIM: Applying basic data cleaning functions: handling missing values using na.omit()/replace_na() in R. import dataset.

>COMMANDS USED

```
# Load necessary libraries
library(dplyr)
library(tidyr) # Contains replace_na()

#
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# 1. SETUP: Create a Sample Dataset with Missing Values (in R)
#
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```

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```
# Creating a small data frame with NA values, using column
names from your file
# to demonstrate the concepts without needing a separate
CSV file.
ai_jobs_sample <- data.frame(
  Job_Title = c("Security Guard", "Research Scientist", "Data
Scientist", "Software Engineer"),
  Average_Salary = c(45795, 133355, NA, 136530),
  Education_Level = c("Master's", NA, "Bachelor's", "PhD"),
  AI_Exposure_Index = c(NA, 0.62, 0.91, 0.39)
)

print("--- 1. Sample Data with NAs (Created in R) ---")
print(ai_jobs_sample)

# Check how many NAs are in each column
print("--- Count of Missing Values per Column ---")
```


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```
print(colSums(is.na(ai_jobs_sample)))

# ---------------------------------------------------------------------------
# 2. METHOD A: REMOVE MISSING VALUES (na.omit)
# ---------------------------------------------------------------------------
# Deletes any row that has at least one missing value (NA).

clean_omit <- na.omit(ai_jobs_sample)

print("--- 2. Data after na.omit() ---")
print(paste("Original rows:", nrow(ai_jobs_sample)))
print(paste("Rows remaining:", nrow(clean_omit)))
print(clean_omit)


# ---------------------------------------------------------------------------
# 3. METHOD B: REPLACE MISSING VALUES
(replace_na)
```

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```
# ------------------------------------------------------------------------
# This is the "surgical option". We fill missing values with
logical defaults.
# Strategy:
# 1. Average_Salary: Fill with the Mean Salary
# 2. Education_Level: Fill with "Not Reported"
# 3. AI_Exposure_Index: Fill missing with 0

# Calculate average salary (ignoring NAs)
avg_salary <- mean(ai_jobs_sample$Average_Salary, na.rm =
TRUE)

clean_replace <- ai_jobs_sample %>%
  replace_na(list(
    Average_Salary = avg_salary,
    Education_Level = "Not Reported",
    AI_Exposure_Index = 0
  ))
```

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```
print("--- 3. Data after replace_na() ---")
print(clean_replace)

# Verify no NAs exist in the columns we cleaned
print("--- Remaining NAs after replacement (Target columns only) ---")
print(colSums(is.na(clean_replace)))
```

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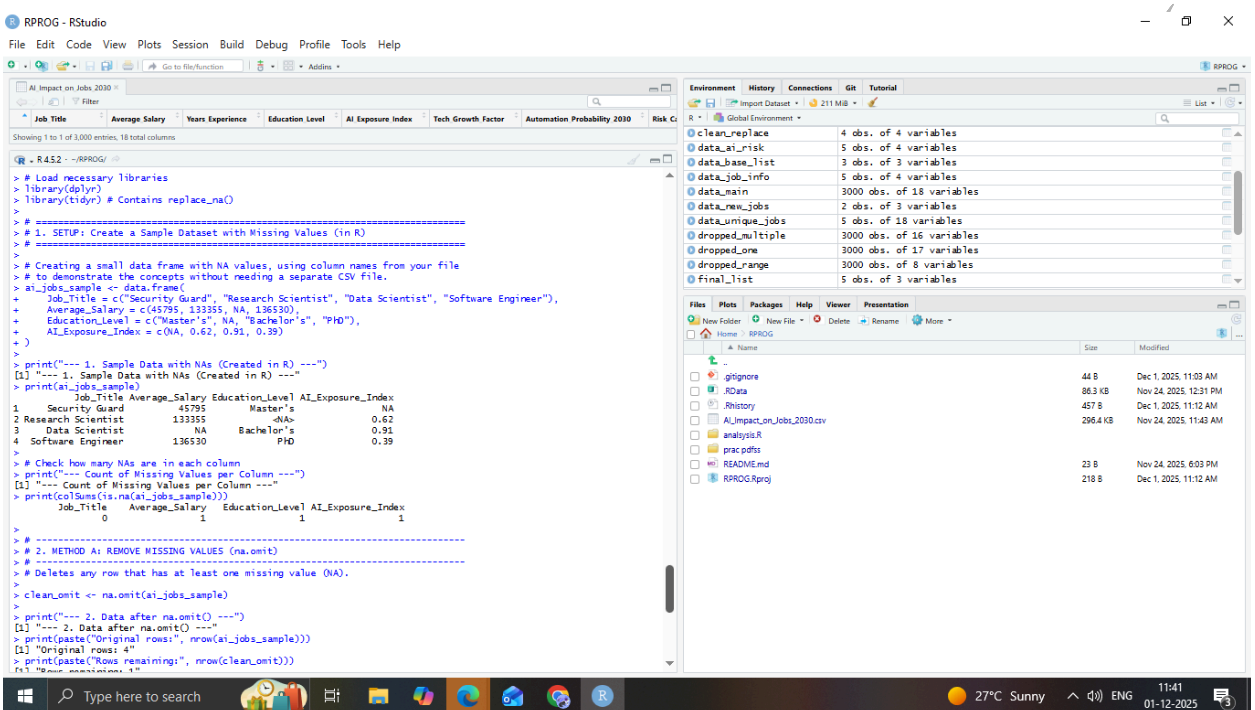

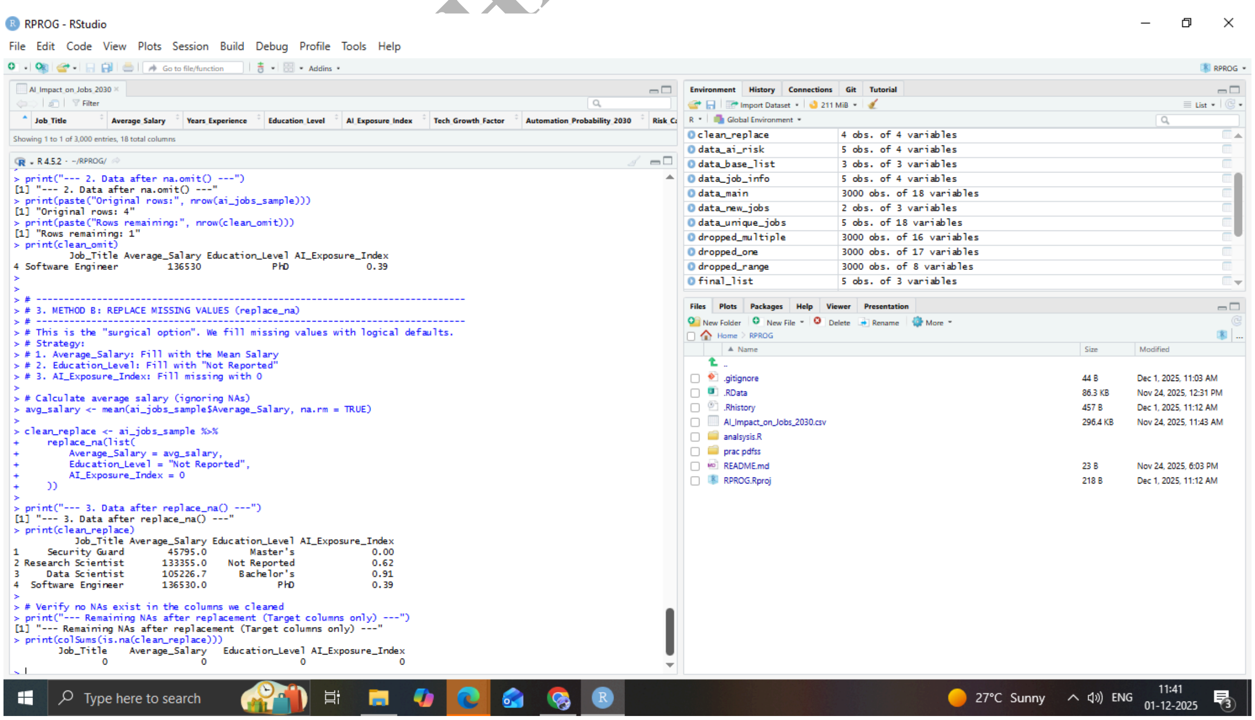

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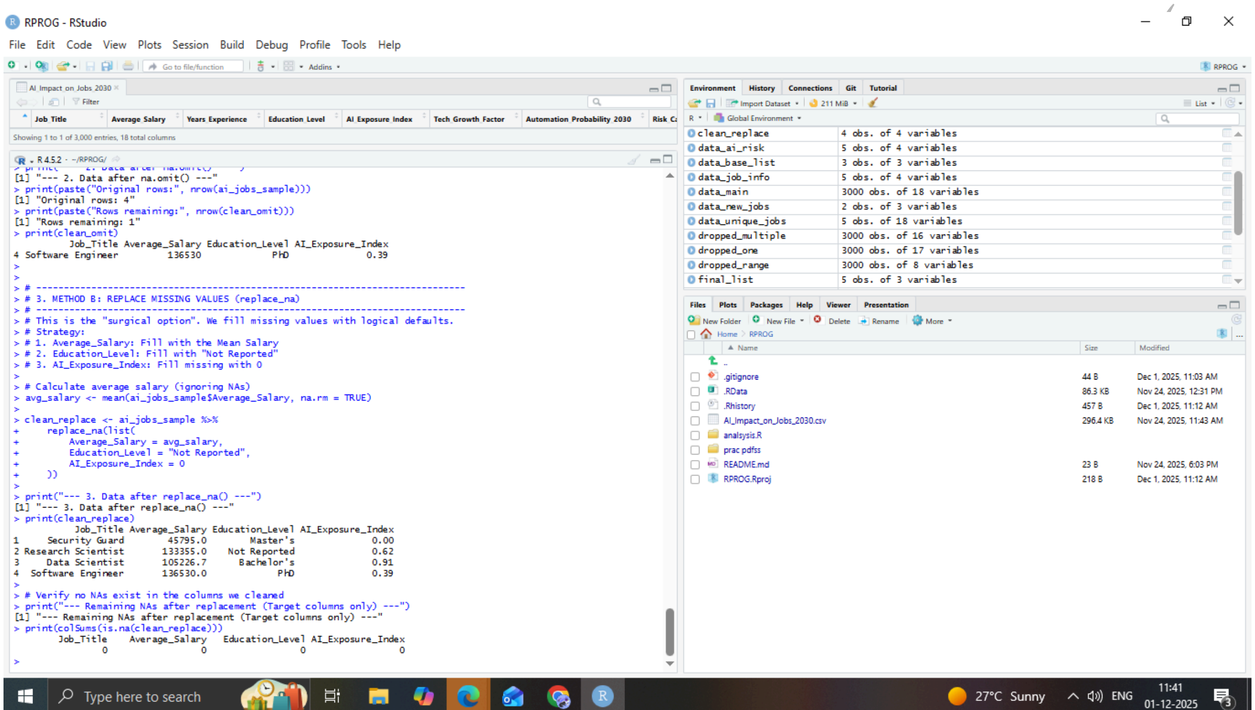

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